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Ablation profile prediction of hard brittle fused silica optics irradiated by single-pulse femtosecond laser

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ABSTRACT

Fused silica optics are extensively utilized in high-power laser facilities due to their superior optical characteristics. Femtosecond laser direct writing is one of the most precision processing techniques that has been used to create functional microstructures on the surfaces of fused silica optics. Complex physical processes occur when femtosecond laser ablates fused silica materials, therefore, precision control of the ablation structure profile during the interaction between femtosecond lasers and fused silica remains a significant challenge. This study addressed this challenge by employing finite element modeling to simulate and analyze the evolution of the ablation profile. Simulations of single-pulse femtosecond laser irradiation were conducted to investigate the impact of laser parameters on the ablation profile. In the model, the evolution of electrons temperature and lattices temperature were revealed according to the two-temperature equation. The ablation temperature isotherm was used to predict the femtosecond laser ablation profile curve. The accuracy of the prediction model was evaluated by comparing the predicted contour curves of various single pulse energies with the experimental results. The finding indicates that the ablation profiles generated by single-pulse femtosecond laser closely match theoretical predictions, with maximum discrepancies in diameter and depth are only 14.36% and 11.07%, and the average deviations of diameter and depth are 6.73% and 7.28%, which verifies the accuracy of the model. With the evolution of single pulse energy, the ablation morphology will appear obvious inner and outer regions and sub-wavelength laser-induced periodic surface structures (LIPSS). The accurate prediction of femtosecond laser ablation profiles is a significant advancement, paving the way for more precise control in laser processing applications.

1. Introduction

Fused silica optics are cornerstone in high-power laser systems, valued for their superior optical and mechanical qualities [1–4]. They are particularly favored for their ability to transmit light in the ultraviolet spectrum, making them ideal for ultraviolet optics applications. However, when exposed to intense ultraviolet laser light, these materials can suffer from initial surface damage that may escalate into more serious laser-induced damage [5–7]. Femtosecond laser has become indispensable in the realm of ultra-precision machining due to their exceptional performance. Femtosecond laser can realize the specific contour processing of materials due to its short pulse width, high power and no heat affected zone. Recent advancements, such as the method

proposed by Fang et al. [8], involve using femtosecond laser selective etching to repair damaged fused silica optics. This technique shapes a three-dimensional contour around the damaged area, which is then removed by an etching solution, effectively restoring the optics integrity. Similarly, Zhou et al. [9] have demonstrated that femtosecond laser can be used to repair damage sites, enhancing the Laser-Induced Damage Threshold (LIDT) of the optics. The application of femtosecond laser has brought new methods to the field of optics processing. Femtosecond laser processing is widely used in optics processing.

The interaction between femtosecond laser and fused silica has been studied by many scholars. Mao et al. [10] have shed light on the ultrafast absorption kinetics of this material under femtosecond laser pulses. Jia et al. [11] have theoretically dissected the damage mechanisms,

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highlighting the critical stages of conduction band electron generation and subsequent laser energy deposition, which significantly influence the ablation process. To gain a comprehensive understanding of experimental observations such as ablation depth and dynamics, these processes are crucial to consider. Xu et al. [12] have furthered this understanding by examining the ablation and damage induced by femtosecond laser in both air and vacuum, revealing distinct behaviors at varying energy levels. When subjected to femtosecond laser irradiation, fused silica experiences nonlinear absorption processes such as multiphoton ionization and avalanche ionization, leading to the generation of free electrons and self-trapped excitons (STEs). These transient electron dynamics play a pivotal role in the laser-induced breakdown and subsequent material modification [13]. The femtosecond laser pulses, with their ultrashort duration and high peak intensity, induce structural changes in the fused silica, such as an increase in the proportion of three- and four-membered ring structures, which results in a densified material state compared to the unirradiated area [14]. This densification enhances the reactivity of oxygen atoms and contributes to the high etching rate observed in femtosecond laser-assisted chemical etching processes [15]. The internal mechanism of the interaction between femtosecond laser and fused silica is one of the significant topics in research, and it is also an important research content for the ablation profile prediction of fused silica materials processed by femtosecond laser.

The application of femtosecond laser extends beyond fused silica, with studies like those by Chen et al. [16–18] demonstrating their use in creating bionic structures on metals and semiconductors for advanced functionalities. Zhao et al. [19] studied the damage characteristics and removal mechanism of femtosecond laser scanning ablation materials. The effects of laser power and scanning speed on the ablation surface quality were analyzed. Guo et al. [20] carried out line scanning experiments with different femtosecond laser powers, and studied the influence of polarization state on the width and morphology of ablation lines. They had expressed the relational of scanning velocity and line overlap rate on the influence of scallop height are established, and then the expressions are calculated. Ding et al. [21–23] have also shown the effectiveness of femtosecond lasers in processing composite materials for electrodes and energy storage. However, mastering ultra-precision machining with femtosecond laser necessitates a deep understanding of the process parameters and the ablation mechanisms. Numerical analysis of the material ablation temperature can elucidate the link between laser parameters and the characteristics of the structures, forming the foundation of ultra-precision machining. Jiang et al. [24] established a phenomenological laser ablation model, and obtained the evolution law of material removal thickness per unit laser flux during laser ablation of fused silica materials and the evolution of material morphology due to incubation effect.

Laser micro-machining, particularly with femtosecond laser, has emerged as a powerful technique for creating micro-holes in transparent materials. This technology offers distinct advantages over traditional methods, making it crucial to refine various aspects of the process. Paramount among these are the optimization of processing methods, the evolution of hole shapes, and the quality and geometry of the hole walls. Mei et al. [25] compared the creation of thermal barrier micro-holes using high-pulse energy femtosecond laser. By conducting drilling experiments under various conditions, they achieved micro-holes with an impressive aspect ratio of 15.7, which could serve as a benchmark for high aspect ratio micro-hole machining in other applications. Jiang et al. [26] enhanced the aspect ratio and quality of micro-holes through a dual-pulse femtosecond laser technique that adjusts free electron density, demonstrating a significant advancement in micro-hole processing. Zhang et al. [27] investigated how parameters such as femtosecond laser spot overlap rate and power affect the drilling of SiC/SiC composites. They found the close relationship between femtosecond laser parameters and the surface quality of the drilled holes. The aspect ratio of micropores and the single pulse removal rate are also critical factors that

influence the outcome of the drilling process.

The objective of this study is to develop a theoretical framework for the rapid ablation of fused silica optics by femtosecond laser, with the aim of prediction ablation profile. This will help elucidate the correlation between the characteristics of the ablation structure and the parameters of the laser. This paper focuses on the deviation between the experimental data and the predicted data of femtosecond laser single pulse ablation of fused silica. The ablation prediction model can open a door for femtosecond laser to repair micro-defects on the surface of fused silica optics in high-power laser facilities. The accuracy and morphology control required for each layer during the repair process are stringent. Therefore, it is necessary to use the model to accurately predict the femtosecond laser ablation morphology. At the same time, the model can be extended to the fields of laser drilling, and the micro-holes with better hole wall quality can be obtained by predicting the single pulse ablation morphology. We have utilized two-temperature model to simulate the progression of the ablation profile on fused silica optics under the influence of single-pulse femtosecond laser. In experimental phase, we conduct tests using an infrared femtosecond laser with a pulse duration of 255 fs. The experiments reveal intriguing observations. With the evolution of femtosecond laser single pulse energy, the ablation morphology will appear LIPSS, which mainly appears near the ablation threshold. They are related to the laser polarization direction. To ensure accuracy of the model, atomic force microscopy (AFM), scanning electron microscopy (SEM), and white light interferometry are employed to characterize the ablation outcomes from the single-pulse femtosecond laser. The morphology characterization results confirm that the experimental findings are in alignment with the theoretical predictions. The prediction results of the ablation contour and the experimental contour results prove the accuracy of the ablation contour prediction model.

2. Model and theory

Fused silica, a type of dielectric material with a wide bandgap, is highly transparent to near-infrared femtosecond laser light. To initiate surface ablation on fused silica using such a laser, a high peak power is necessary. This power level triggers a nonlinear absorption of light, which in turn creates plasma within the material through mechanisms such as single-photon and multiphoton ionization. The simultaneous action of the plasma and the laser is what leads to ablation. The process of plasma generation starts with the excitation of electrons from the valence band to the conduction band, creating a small number of seed electrons. These electrons then initiate a chain reaction through collision ionization, where a significant amount of energy is deposited, eventually causing the material to ablate. The decision between multiphoton ionization and collision ionization during the interaction is guided by the Keldysh parameter, which indicates the dominant process [28–30]. When a near-infrared femtosecond laser interacts with the surface of fused silica, it rapidly generates a considerable number of free electrons within the material. Despite being less dense than in conductors and metals, these free electrons are capable of absorbing the laser energy within the skin depth range. Once energized, these electrons transfer their energy to the material lattice system through a process known as photoacoustic coupling. Given the brevity of the photoelectric and photoacoustic phases, traditional Fourier heat transfer models are not well-suited to describe this energy transfer in detail. In this study, we use a two-temperature model to accurately depict the rapid energy transfer process, where the electrons action time is significantly shorter than the interaction time between the electrons and the lattice [31,32]. The focus of our theoretical analysis is on the dramatic energy changes within the electron system, with the energy consumption during the phonon action process considered negligible. Furthermore, we assume that there is no energy exchange occurring between the fused silica and its surrounding environment during the laser interaction process. This simplification allows us to concentrate on the intrinsic material response to the laser, providing a clear understanding of the ablation and melting dynamics.

The energy coupling model can be represented as [33]:

$$C_e \frac{\partial T_e}{\partial t} = \frac{\partial}{\partial z} \left(k_e \frac{\partial T_e}{\partial z} \right) - g(T_e - T_l) + I_{eff} \quad (1)$$

$$C_l \frac{\partial T_l}{\partial t} = g(T_e - T_l) \quad (2)$$

where, C_e represents the electronic heat capacity, C_l is the lattice heat capacity, T_e is the electronic temperature, T_l is the lattice temperature, K_e is the electronic thermal conductivity, g is the electron-phonon coupling coefficient, and I_{eff} is the laser source term.

When femtosecond laser interacts with a material, the focal plane of the femtosecond laser occurs at different depths within the material, causing variations in the material absorption depth for the laser. In the theoretical research conducted in this study, it is assumed that the femtosecond laser is vertically incident on the surface of the fused silica optics. Therefore, the equivalent laser heat source term can be expressed as:

$$z \quad (3)$$

where, I_{peak} represents the peak power density, f_{space} denotes the spatial distribution function, and f_{time} represents the temporal distribution function. Typically, femtosecond laser exhibit Gaussian distributions in both spatial and temporal domains, and I_{peak} , f_{space} , and f_{time} can be expressed as:

$$I_{peak} = \frac{2F}{\sqrt{\pi/\ln 2} \cdot t_p} \quad (4)$$

$$f_{space} = \exp[-2(\omega_s/\omega)^2] \quad (5)$$

$$f_{time} = \exp[-(4\ln 2) \cdot (\frac{t}{t_p})^2] \quad (6)$$

where, F represents the energy density, t_p denotes the pulse width, ω represents the waist radius, and ω_s represents the distance from the focal spot center. In the simulation of femtosecond laser in this study, considering that a portion of the energy is absorbed and another part is scattered into the environment, according to Lambert-Beer law, the absorbed laser energy can be expressed as:

$$I_{ab} = I_{total} \cdot [1 - R(t, \omega_s)] \cdot \alpha(t, \omega_s, 0) \cdot \exp(-\int_0^z \alpha(t, \omega_s, z) dz) \quad (7)$$

where, $\alpha(t, \omega_s, Z)$ represents the absorption depth, $R(t, \omega_s)$ represents the reflectance, and z represents the depth from the material surface along the direction of laser propagation. The final femtosecond laser heat source term can be expressed as:

$$I_{eff} = \frac{2F}{\sqrt{\pi/\ln 2} \cdot t_p} [1 - R(t, \omega_s)] \cdot \alpha(t, \omega_s, 0) \exp[-\frac{\omega_s^2}{\omega^2} - (4\ln 2) \cdot (\frac{t}{t_p})^2 - \int_0^z \alpha(t, \omega_s, z) dz] \quad (8)$$

During multiphoton ionization, there is a rapid increase in the free electron density within the material, which then levels off at a critical value. Our paper examines the energy exchange between electrons and the material on a picosecond timescale, which is after the free electron density has reached its critical level. In this model, the laser source term I_{eff} accounts for the energy transfer process that takes place once the free electron density has stabilized. Combining Equation (8) where femtosecond laser energy spatially follows a Gaussian distribution, the finite element method is employed for solving. The computational domain is set with a width of 500 μm and a height of 50 μm , and the minimum grid size is 0.3 $\mu\text{m} \times 0.3 \mu\text{m}$. The simulation parameters used in the process are listed in Table 1.

In the simulation process, room temperature is set as the initial

Table 1

Thermodynamic parameters of fused silica materials.

Items	Symbol	Values
Lattice hot melt	C_l	1.549E6 J/m ³ /K [34]
Coupling coefficient	g	3E16 W/m ³ /K [34,35]
Electronic specific heat Capacity coefficient	C_{eo}	1E-16 J/um ³ /K ² [36]
Electron heat capacity	C_e	$C_{eo} \times T_e$ [34]
Electron thermal conductivity	K_{eo}	$K_e \times (T_e/T_l)$ [36]
Lattice thermal conductivity	K_l	$0.5 + 2.9E-4 \times (T_l - T_{glass}) \times 10E-4$ [36]
Melting point of glass	T_{glass}	1873.2 K
Reflectivity	S	0.03[37]
Absorption depth	a	3E6 1/m [38]
Energy Density	I_0	2.4E5 J/m ²
Pulse width	t_p	255 fs
Spot radius	R_0	3 μm

condition both inside the material and at the edges, while the edges beyond the laser incident boundary are set as adiabatic. Equations (1) and (2) are coefficient-type partial differential equations, typically solved using finite difference methods. The finite difference format can be expressed as:

$$C_e \left(\frac{T_{ij+1}^e - T_{ij}^e}{\Delta t} \right) = \frac{1}{\Delta z} \left(\frac{T_{ij+1}^e - T_{ij}^e}{\Delta z} \right) k_e - g(T_{ij+1}^e - T_{ij+1}^l) + I_{eff}(\Delta z) \quad (9)$$

$$C_l \left(\frac{T_{ij+1}^l - T_{ij}^l}{\Delta t} \right) = g(T_{ij+1}^e - T_{ij+1}^l) \quad (10)$$

The modeling process described above simplifies the intricate interaction between femtosecond laser and fused silica material through finite element modeling, as shown in Fig. 1. This approach provides a visual representation of the thermodynamic characteristics at any location within the material. In the model, the fused silica optic is a cylinder with a radius of 50 mm and a height of 5 mm, and the minimum mesh size is 50 nm $\times$ 100 nm.

3. Experimental design

The corresponding femtosecond laser ablation experiments are carried out to verify the accuracy of the ablation profiles obtained from the theoretical model. The aim is to investigate the mapping relationship between the femtosecond laser processing parameters and the surface diameter and depth of the fused silica material. In conjunction with the theoretical findings, the experiments are designed to test the accuracy of the predicted femtosecond laser ablation profiles of fused silica optics.

The femtosecond laser path is illustrated in Fig. 2. This chapter outlines the experimental design, including a single-point ablation experiment employing single-pulse femtosecond laser. The femtosecond laser energy ranges from 5 to 12 μJ , with 10 measurements taken at each energy level to analyze the depth and diameter of the ablated points. In this work, the wavelength of femtosecond laser is 1030 nm, the pulse width is 255 fs, and the spot diameter after focusing is 6 μm . According to Equation (11), the diameter of the laser spot focused by the femtosecond laser theory can be calculated. Considering that the femtosecond laser is interfered by many factors during the transmission process, the beam quality M^2 that eventually reaches the focusing objective lens cannot be accurately determined. Therefore, the relationship between the diameter D of the ablation area and the pulse energy E_p is established, as shown in Equation (11):

$$D^2 = 2\omega^2 \ln \left(\frac{2E_p}{\pi\omega^2 F_{th}} \right) \quad (11)$$

where ω is the focused laser spot radius, D is ablative zone diameter (5.8 μm), E_p is pulse energy (8 μJ), and F_{th} is the ablation threshold of the material (8.5 J/cm²). We used the femtosecond laser repetition frequency was set to 1 kHz, the scanning speed was maintained at 50 mm/s,

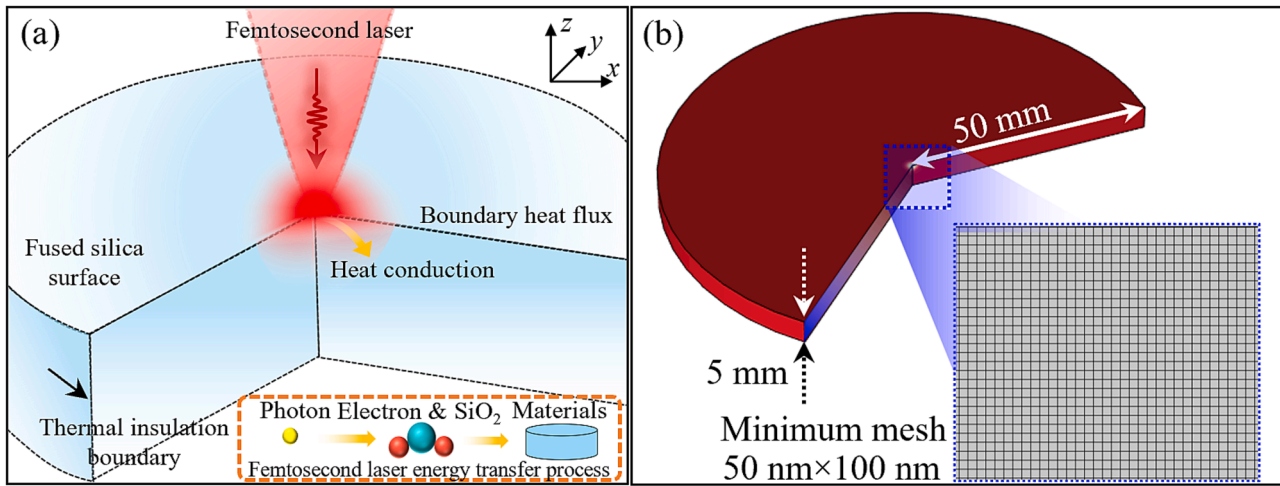


Fig. 1. Finite element simulation model. (a) The model of femtosecond laser irradiation with fused silica material established to reveal the laser energy transfer process on basis of the two-temperature equation. The boundary conditions are set so that there is no energy exchange between the material and the environmental system; (b) The fused silica optic is a cylinder with a radius of 50 mm and a height of 5 mm, and the minimum mesh size is 50 nm $\times$ 100 nm.

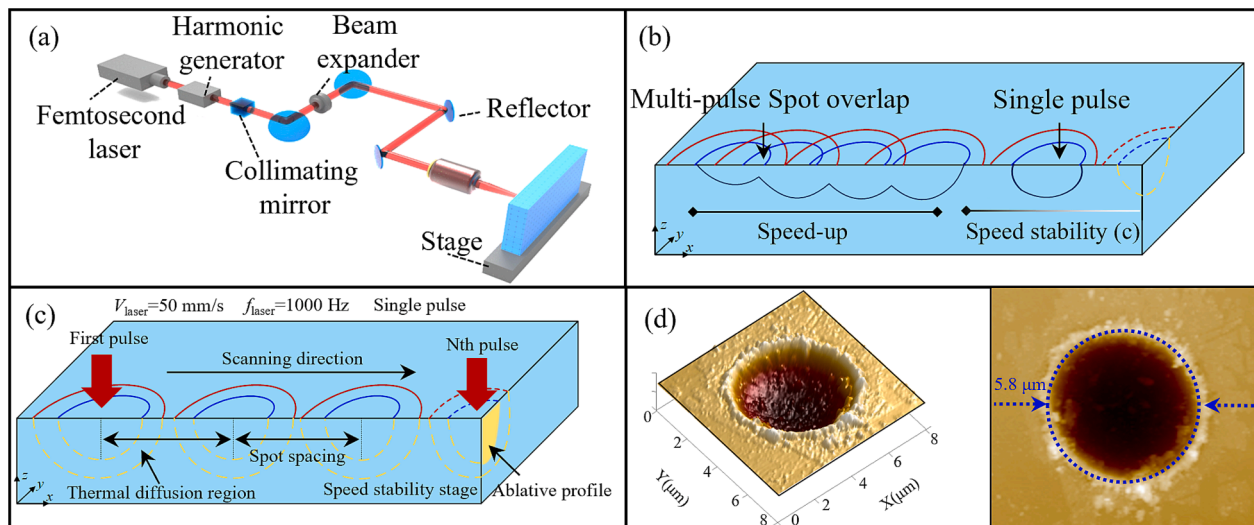


Fig. 2. Femtosecond laser optical path and experimental design of femtosecond laser processing. (a) Femtosecond laser experimental optical path diagram; (b) The femtosecond laser ablates the fused silica material at 50 mm/s, which includes a speed increase phase and a speed stabilization phase; (c) In the stable stage of femtosecond laser scanning speed, single pulse ablation of fused silica material will lead to obvious regional boundary caused by energy transfer in the single pulse ablation area; (d) AFM measurement of single pulse femtosecond laser ablation diameter.

and the scanning length was extended to over 5 cm. These adjustments allowed for the creation of a single pulse laser ablation crater on the surface of fused silica, which is crucial for accurate measurement. The focused spot diameter has been measured using AFM and the diameter of ablation area was calibrated by three-point fixed circle method, as shown in Fig. 2 (d). In the initial stage of scanning, the moving speed of the platform gradually increases, and there is overlap between multiple femtosecond pulses. When the platform speed is stable, a single pulse processing fused silica element can be observed.

4. Results and discussion

4.1. The temperature evolution of femtosecond laser interaction with fused silica optics

Simulations provide insights into the response of fused silica material to single-pulse laser irradiation, focusing on the electron and lattice system temperatures, as shown in Fig. 3, three positions in the Z

direction and two positions in the X direction are selected respectively in the simulation model of the single-pulse irradiated fused silica material to obtain their electron and lattice temperatures, as shown in Fig. 3. The laser parameters used in the simulation are consistent with Table 1. The single pulse energy is 8 μ J and the spot radius is 3 μ m.

Initially, as depicted in Fig. 4 (a1) and (a2), the material electron temperature spikes to a peak of 82,000 K within a mere 260 fs upon laser exposure, while the lattice temperature is still in the process of increasing. After a relaxation period of several hundred picoseconds, the lattice temperature peaks at 43,000 K, after which the electron temperature drops rapidly, and the lattice temperature decreases more gradually, with both converging at a lower temperature. This observation suggests that following the femtosecond laser pulse, the relaxation times for both the electron and lattice systems in fused silica are on the order of hundreds of picoseconds. During this period, the material is subjected to extremely high temperatures. Once the temperatures stabilize, the internal temperature of the material remains significantly below the threshold for ablation, which is 3,000 K. This indicates that

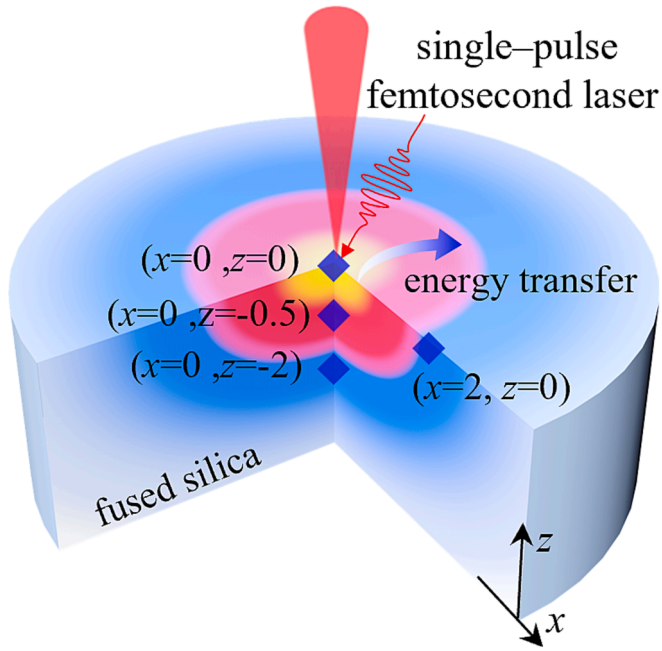


Fig. 3. The schematic diagram of the observation position selected in the simulation process.

the energy imparted by the femtosecond laser is confined to a relatively shallow depth within the material, preventing widespread thermal effects and ensuring localized processing. This controlled energy deposition is crucial for achieving precise modifications to the fused silica material without causing extensive thermal damage, a key advantage of femtosecond laser processing.

We focus on the temperature dynamics at the surface and at varying depths within the fused silica material when subjected to femtosecond laser irradiation. At the surface ($z = 0 \mu\text{m}$), the electron temperature peaks at an extraordinary 82635 K within a mere 260 fs, highlighting the rapid energy absorption capacity of the material. Following this initial spike, it takes approximately 4.4 ns for the electron and lattice systems to reach thermal equilibrium, settling at a final equilibrium temperature of around 18252 K. As we delve deeper into the material, at a depth of $0.5 \mu\text{m}$, the electron and lattice temperatures still require a relaxation period to synchronize, after which they decrease concurrently. At a

depth of $2 \mu\text{m}$, the temperatures of the electron and lattice systems remain consistent, indicating that the temperature rise at this depth is not a direct result of laser energy absorption but rather due to thermal diffusion from the surface. This phenomenon is not observed in the surface temperature transfer, as depicted in Fig. 4 (d1) and (d2), which suggests that energy transfer into the material interior is less efficient compared to the surface, where laser effects are more pronounced. The rapid increase and subsequent decrease of electron temperature, while the lattice system temperature remains relatively lower initially, creates an energy imbalance. This imbalance drives the transfer of energy from the electrons to the lattice, causing the electron temperature to drop and the lattice temperature to rise until they reach an equilibrium state.

In this study, we examine the rapid thermal response of fused silica optics when subjected to a single pulse from femtosecond laser with an energy of $8 \mu\text{J}$, the material lattice temperature distribution of $180 \text{ ps} \sim 50 \text{ ns}$ by single-pulse femtosecond laser ablation of fused silica material is shown in Fig. 5. We are concerned with the temperature change in the lattice system in a relatively short period of time after femtosecond laser action on fused silica materials. This temperature rise culminates at a peak 180 ps after the laser impact. Post this peak, the lattice system initiates a cooling process, with energy dissipating into the material. By the 10 ns mark, the material undergoes maximum ablation, as depicted in the accompanying Fig. 5. This Fig. 5 illustrates that the ablation contour expands progressively from 180 ps to 10 ns . Beyond 10 ns , there will be no ablation of the fused silica material. In the context of femtosecond laser processing, the pulse frequency typically far exceeds the pulse width, indicating that the processing effect is an accumulation of multiple single pulses. Consequently, the heat-affected zone in such processes remains minimal, ensuring precision and control over the material modification.

To study and validate the established simulation model, theoretical ablation profiles and the depth and diameter of micropores were obtained. Single-pulse ablation energy in the range of 4 to $11 \mu\text{J}$ is chosen for the following validation experiments. Based on the mechanism of phase explosion during femtosecond laser ablation, transitioning from normal vaporization to phase explosion, the conditions for femtosecond laser ablation are set as the material ablation temperature of fused silica at 3000 K . As shown in Fig. 6, the predicted ablation profile is hyperbolic in X direction, and the ablated micropore exhibits a bowl-shaped shallow pit.

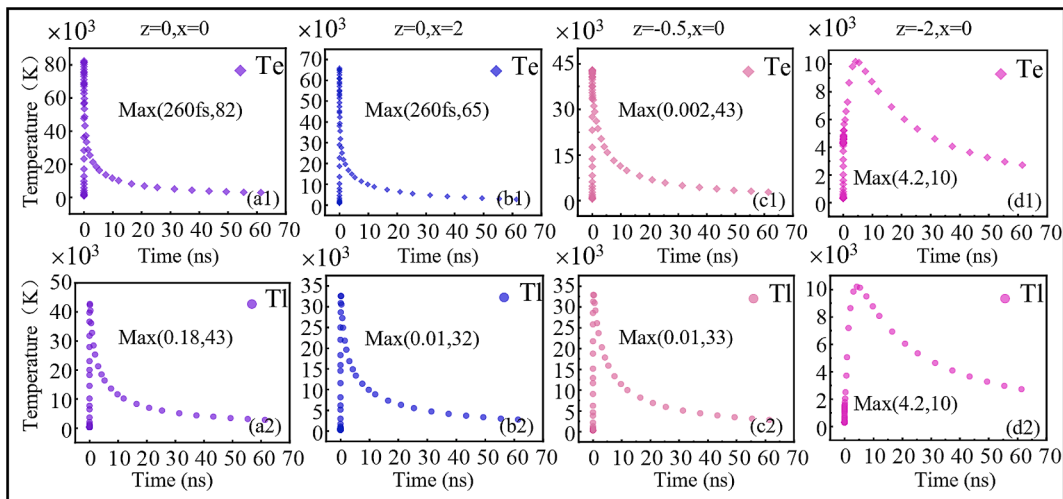


Fig. 4. Curves of electron temperature and lattice temperature at different spatial positions. (a1), (a2) The electron temperature (T_e) and lattice temperature (T_l) curves of the original point position in the simulation model; (b1), (b2) The T_e and T_l curves of the surface $2 \mu\text{m}$ away from the origin; (c1), (c2) The T_e and T_l curves with a depth of $0.5 \mu\text{m}$ from the surface; (d1), (d2) The T_e and T_l curves with a depth of $2 \mu\text{m}$ from the surface.

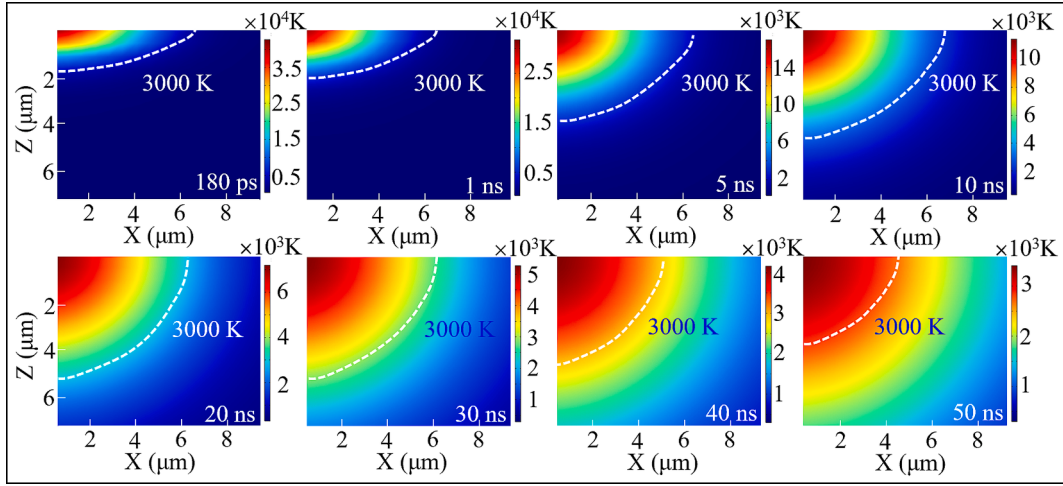


Fig. 5. The material lattice temperature distribution of 180 ps ~ 50 ns, and the white dashed line represents the ablation temperature of fused silica material. This model accounts for the distinct temperatures of electrons and the lattice, reflecting the different rates at which they absorb and dissipate energy. The material quickly reaches a high temperature in a very short time. At this time, the temperature exceeds the melting point of the material, and the local high temperature cannot diffuse in a short time, so the local instantaneous high temperature and ‘cold’ processing are not in conflict. In order to eliminate the confusion of readers, the ‘cold’ processing is described in this paper.

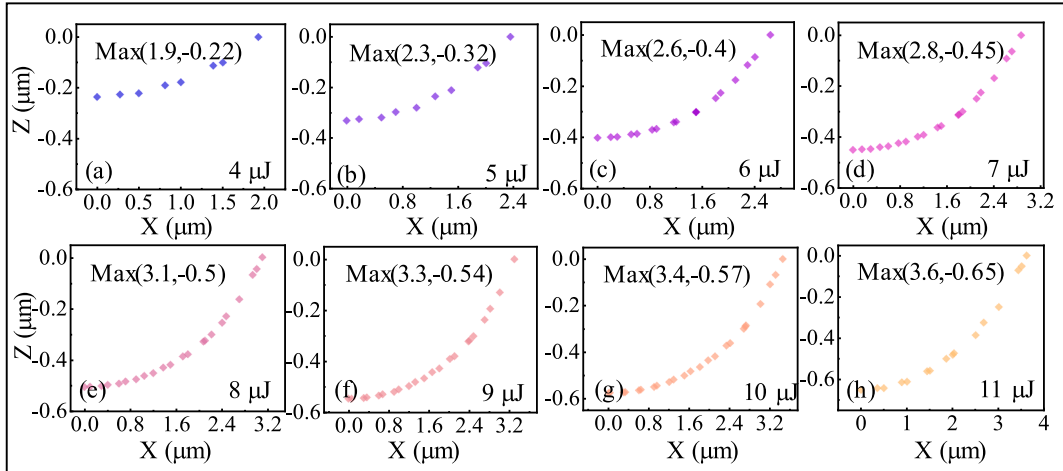


Fig. 6. 4 ~ 11 μJ simulated contour curve of fused silica irradiated by femtosecond laser. (a)~(h) The ablation profiles of fused silica optics obtained by a single-pulse femtosecond laser at 4 ~ 11 μJ are predicted.

4.2. Morphology evolution with single pulse energy

The experimental investigation commenced by studying the ablation outcomes using femtosecond laser single-pulse irradiation. In these

experiments, a femtosecond laser with a fixed wavelength of 1030 nm and a pulse width of 255 fs was focused onto the surface of fused silica material, resulting in a laser beam spot diameter of 6 μm . Single-pulse ablation was conducted on the surface of the fused silica optics by

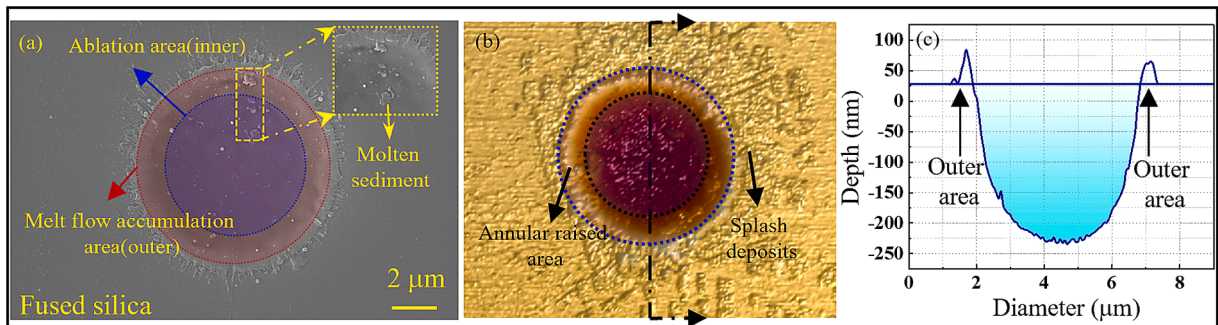


Fig. 7. SEM and AFM images of fused silica optics irradiated by single-pulse femtosecond laser. (a) Single-pulse irradiation on the surface of fused silica optics forms a ring structure with distinct inside and outside. (b) AFM images of single-pulse irradiation. (c) Single-pulse irradiation will form an internal area and an external area, and the height of the external area can be clearly observed.

controlling the scanning frequency and scanning speed, generating multiple micropores with varying single-pulse energies. SEM and AFM were employed to observe the ablated micropores, and the final diameters and depths were obtained as the average values from ten ablated micropores. The comparative diagram in Fig. 7, AFM and SEM results, allows for a clear distinction between the inner and outer rings of the femtosecond laser single-pulse ablated micropores. The inner ring represents the region where ablation occurs, while the outer ring delineates the boundary of the entire treatment area. External to the outer ring, some splatter particles are visible, formed by particles splashing back onto the material surface.

In the conducted experiments, the effects of varying single-pulse energies on the ablation characteristics of fused silica optics were meticulously examined. At the lower end of the energy, the ablation craters were observed to be nearly circular and shallow, indicating a more uniform and less intense interaction with the material. Notably, the absence of melted material within the ablation points suggests that the energy levels were insufficient to cause melting at these stages. As the energy input increased to intermediate levels, the formation of re-deposited material within the ablation region became evident. This was accompanied by the appearance of spattered material around the periphery of the ablation zone. Energy Dispersive Spectroscopy (EDS) analysis confirmed that the spattered material was composed of SiO_2 , confirming that it originated from the fused silica material that had melted and was ejected due to the high energy of the laser pulses. As shown in the Fig. 8, irradiation of the fused silica material by a single-pulse femtosecond laser results in bumps of different heights depending on the energy. The small increase in the depth of the ablated

structure is due to the fact that the femtosecond laser irradiation of fused silica material results in the formation of a protective plasma film on the ablated surface, which leads to no significant increase in the depth of the ablation.

In order to investigate the change of the ablation profile of fused silica materials by femtosecond laser ablation, we conducted experiments to determine the relationship between the diameter of the ablated micropores and the energy of a single-pulse laser. The experiments utilized single-pulse ablation with a femtosecond laser, as illustrated in Fig. 9. Our observations indicate a linear relationship between the diameter of the micropores and the single-pulse energy. A comparison of the ablation effects at energies of 5 μJ and 10 μJ revealed a significant difference in the extent of ablation, with a marked difference in pore diameter. Additionally, periodic structures within the micropores were observed during ablation at 9 μJ , a phenomenon that will be discussed in more detail in later sections of this paper. Continuous monitoring of the ablated fused silica diameter as the single-pulse energy is increased from 1 μJ to 10 μJ showed no significant changes, suggesting a consistent energy output from the laser spot.

In parallel experiments examining the relationship between the depth of the ablated micropores and the single-pulse energy, a linear correlation was again observed, as depicted in Fig. 9. However, the increase in depth from 0.22 μm to 0.53 μm as the energy rose from 1 μJ to 10 μJ was not substantial. It was found that a single-pulse energy exceeding 0.7 μJ could initiate ablation on the fused silica surface. Despite this, the depth of ablation did not increase significantly with the rise in energy from 1 μJ to 10 μJ in single-pulse experiments. The primary reason for this outcome is the extremely brief interaction time of

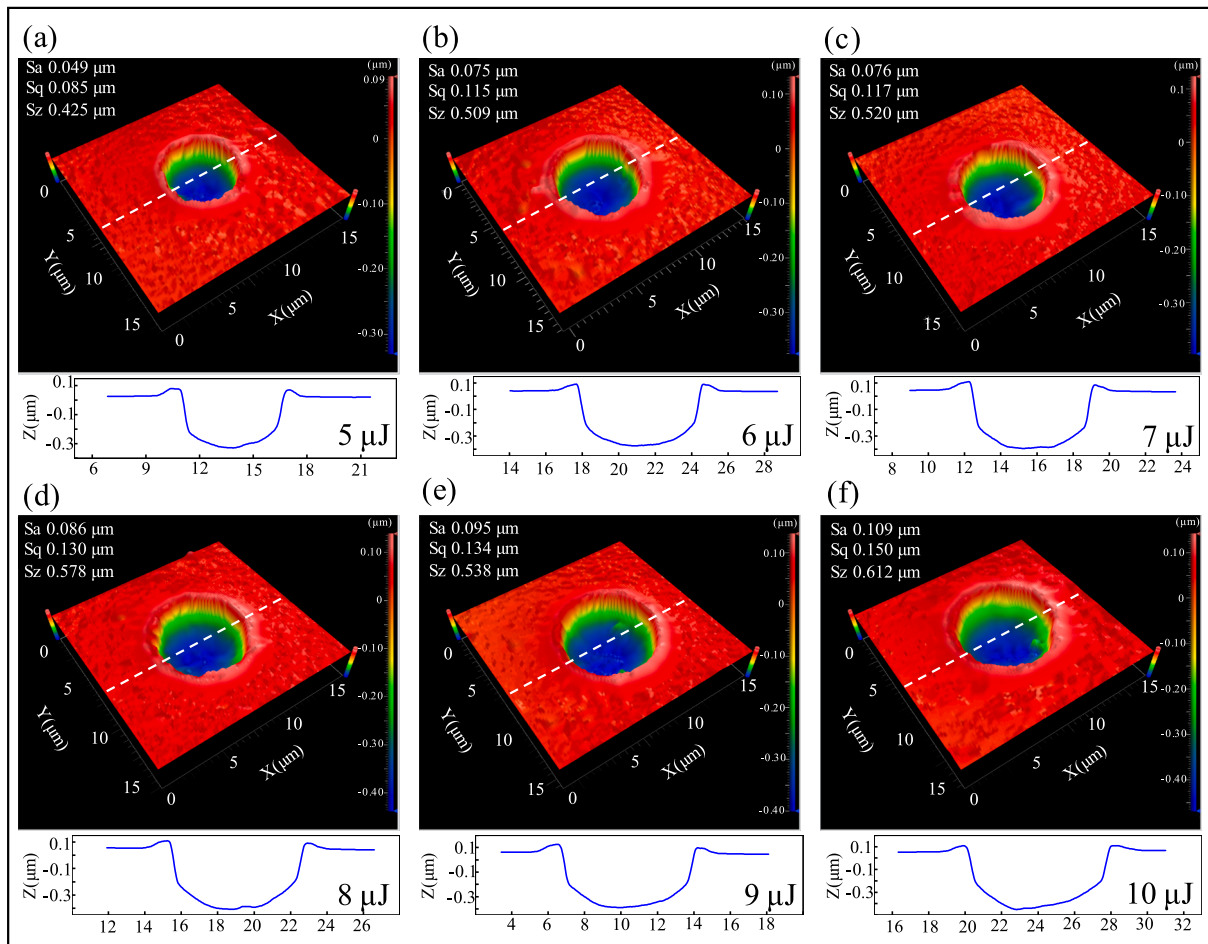


Fig. 8. The experimental surface morphology of fused silica is ablated by femtosecond laser when the single pulse energy is 5 ~ 10 μJ . (a)~(f) The diameter and depth of the single-pulse ablation structure gradually increase with the increase of single-pulse energy, and the single-pulse energy increases from 5 μJ to 10 μJ .

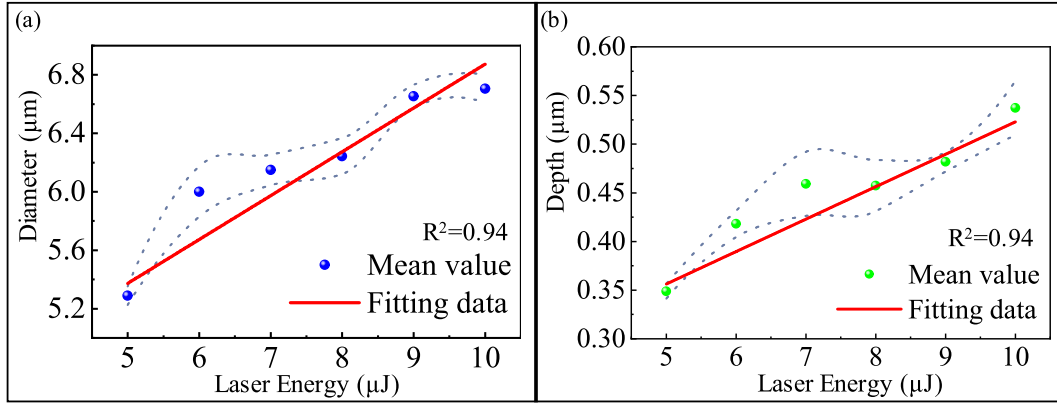


Fig. 9. The change of micro-pit diameter and depth of fused silica irradiated by femtosecond laser with single-pulse energy of 5 ~ 10 μJ. (a) Diameter of the ablative profile; (b) Depth of the ablative profile.

the femtosecond laser with the material, which precludes energy accumulation during a single pulse. At the same time, due to the high density of plasma generated during femtosecond laser irradiation of fused silica materials, a protective film is formed on the surface of the ablated structure, resulting in insignificant changes in the depth of ablation.

These findings underscore the precision of femtosecond laser ablation, where energy is deposited in a controlled manner within the material surface layer. The experiments highlight the potential for fine-tuning ablation characteristics by adjusting pulse energy, while also indicating the limitations of depth penetration with single-pulse irradiation. This understanding is crucial for optimizing the process parameters in applications requiring precise micromachining of fused silica and other transparent dielectric materials.

The presence of plasma during the ablation process introduces additional complexities. The plasma, generated by the laser interaction with the material, cannot be entirely removed through splashing and melting, leading to traces of re-deposition within the ablation points. Fig. 10 provides a clear illustration of this phenomenon, showing a significantly deepened ablation point under conditions of high energy.

The pressure exerted by the plasma explosion drives more melted material to flow both downward and outward, as depicted in Fig. 10. This results in the formation of periodic structures within the central region of the ablation point, which can adversely affect the quality of the processing. Notably, when the single-pulse energy surpasses 9 μJ, a pronounced periodic structure emerges in the inner region of the ablation point. The orientation of the stripe pattern within this structure is likely influenced by the polarization direction of the laser, exhibiting a circular arrangement. This detailed examination of the ablation process under varying energy conditions contributes to a deeper understanding of the physical mechanisms at play during femtosecond laser ablation and melting of fused silica optics. The findings are instrumental in optimizing laser processing techniques and advancing the development of high-precision laser applications.

When the incident single-pulse energy approaches the material damage threshold, different optical absorption effects occur in the focal point of the ablation and in the surrounding material. Due to the high laser energy density at the center of the ablation focus, nonlinear absorption effects stimulate more electrons, while the periphery of the

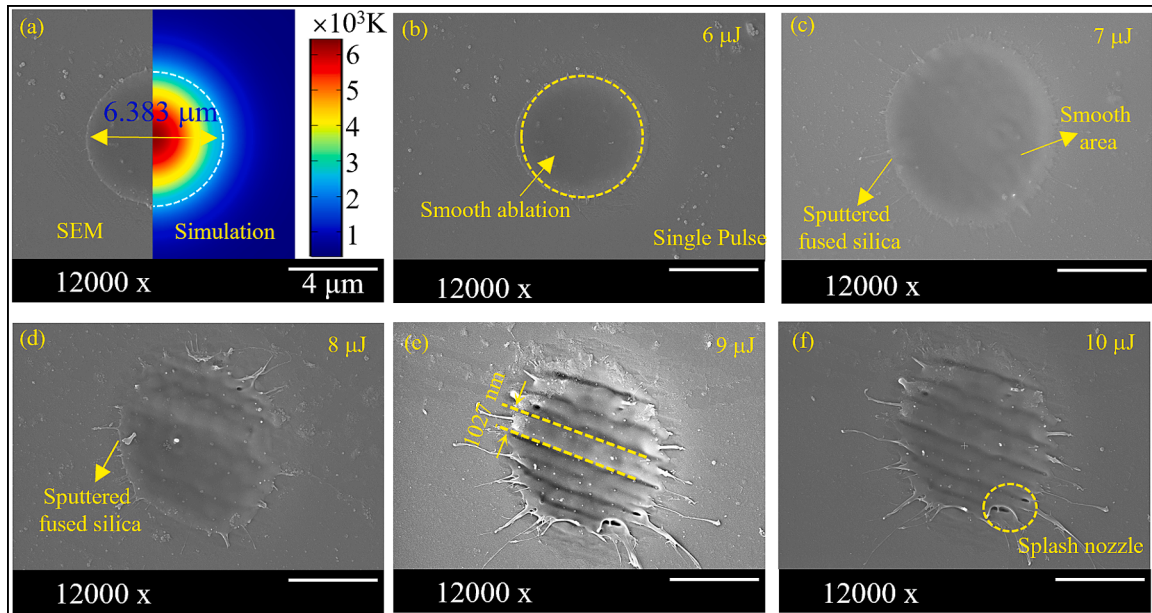


Fig. 10. SEM image of fused silica material irradiated by single-pulse femtosecond laser. (a) The results of SEM observation are compared with the simulation results, the single-pulse energy is 6 μJ. (b) Shallow pits with the same spot appear when the single-pulse energy is 6 μJ; (c) Single-pulse action diagram with distinct internal and external regions when the single-pulse energy is 7 μJ. (d ~ f) LIPSS related to laser energy and polarization direction generated during single-pulse irradiation. When the single-pulse energy is near 9 μJ, subwavelength LIPSS are produced.

laser-irradiated region is influenced by thermal diffusion. As the pulse energy increases to 6 μJ , thermal effects become significant, and the influence of the molten fluid motion becomes increasingly pronounced, leading to a decline in ablation quality. Moreover, at high single-pulse energy states, the presence of spalled particles generated by the jetting process forms craters at the edges of the ablation point, and a large amount of redeposited material splashes around the ablation region.

4.3. Model validation and morphology prediction

The comparison between the ablation profile predicted by the model and the contour curve obtained by the experiment shows that the diameter and depth errors of the ablation structure predicted by the model are small. As shown in Fig. 11 (a) ~ (b), the diameter and depth of the predicted ablation profile are similar to the curves obtained by the experiment. This study finds that, the simulated results of the diameter and depth of ablated micropores for 5 to 10 μJ are consistent with experimental results. The error between the ablation diameter, depth, and theoretical values is shown in Table 2, the minimum error is 1.14 % and 4.72 %. For the diameter of the ablation profile, when the single pulse energy is 8 μJ , the deviation between the experimental data and the model prediction data is large, and the maximum error is 14.36 %. The reason for this situation is that the boundary between the inner and outer regions of the ablation structure is not obvious enough in the process of single-pulse ablation of materials, and the diameter of the ablation region cannot be accurately measured during the measurement process, so there is a certain difference with the theoretical prediction value. For the depth, when the single pulse energy is between 7 ~ 10 μJ , the maximum error between the experimental data and the model prediction data is 11.07 %, and the experimental data are smaller than the theoretical prediction data. The reason for this situation is that the plasma melts and deposits at the bottom of the ablation structure during the ablation process, resulting in a decrease in depth. At the same time, the formation of the LIPSS structure will further increase the error of the prediction results.

5. Conclusion

Femtosecond laser ultra-precision machining is one of the key technologies for preparing functional structures on fused silica optics. The interaction between femtosecond laser and fused silica material contains complex physical processes, making it a major challenge to predict the ablation morphology of femtosecond laser. Ablation profile prediction is very important for femtosecond laser repairing micro-defects on the surface of fused silica optics, which can effectively improve the surface quality of femtosecond laser repaired structures. This study delves into the thermal dynamics of fused silica when

Table 2

Comparison of experiments and simulations on single-pulse femtosecond laser ablation of fused silica material.

Energy (μJ)	Diameter (μm)	Deviation	Depth (μm)	Deviation
5	5.289	14.36 %	0.351	6.72 %
6	5.997	13.88 %	0.417	5.04 %
7	6.105	6.88 %	0.424	4.72 %
8	6.383	2.98 %	0.452	9.6 %
9	6.653	1.14 %	0.482	11.07 %
10	6.798	1.15 %	0.537	6.51 %

subjected to femtosecond laser irradiation, utilizing the two-temperature model to elucidate the underlying mechanisms. The initial interaction between the laser and the material involves a rapid, nonlinear absorption of energy by the electrons, which is swiftly followed by the coupling with the material lattice structure. This process culminates in the transfer of energy from the lattice to the material interior, resulting in material ablation. For instance, with a single pulse energy of 8 μJ , the temperature within the lattice system can spike within 180 ps. And the material could be removed, yielding the ablation profile. To test the theoretical model accuracy, experimental irradiation of fused silica with a single-pulse femtosecond laser is conducted. The experimental ablation profiles are compared with simulated ones, revealing discrepancies in diameter and depth of only 1.14 % and 4.72 %. The deviation of the model prediction is mainly due to the plasma melting deposition and the formation of LIPSS. This level of accuracy in predicting ablation profiles is a significant step forward, which provide important reference for femtosecond laser precision machining. In the future, we will focus on the multi-pulse, large-scale, full-process femtosecond laser ablation morphology prediction model to carry out in-depth research, making the femtosecond laser ablation morphology prediction more accurate and perfect.

CRediT authorship contribution statement

Tianyuan Li: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Zhaoyang Yin:** Writing – review & editing, Software, Data curation. **Mingjun Chen:** Writing – review & editing, Funding acquisition. **Jian Cheng:** Writing – review & editing, Project administration, Funding acquisition. **Linjie Zhao:** Writing – review & editing, Project administration, Funding acquisition, Formal analysis. **Tianhao Zhang:** Software, Data curation.

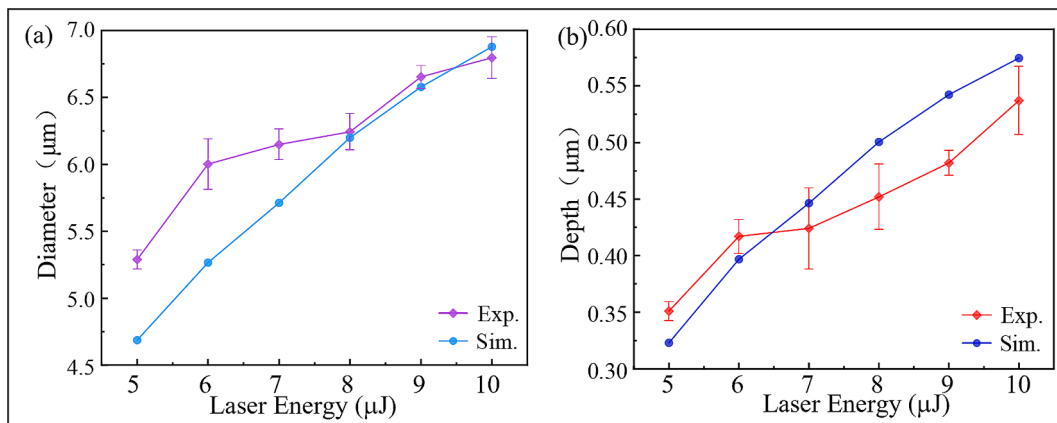


Fig. 11. (a) Diameter of predicted profile and ablative profile; (b) Depth of predicted profile and ablative profile.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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